Zoology Legends

Zoo102

Quiz No.3 Spring 2023

Muhammad Zaman

0304-4756496

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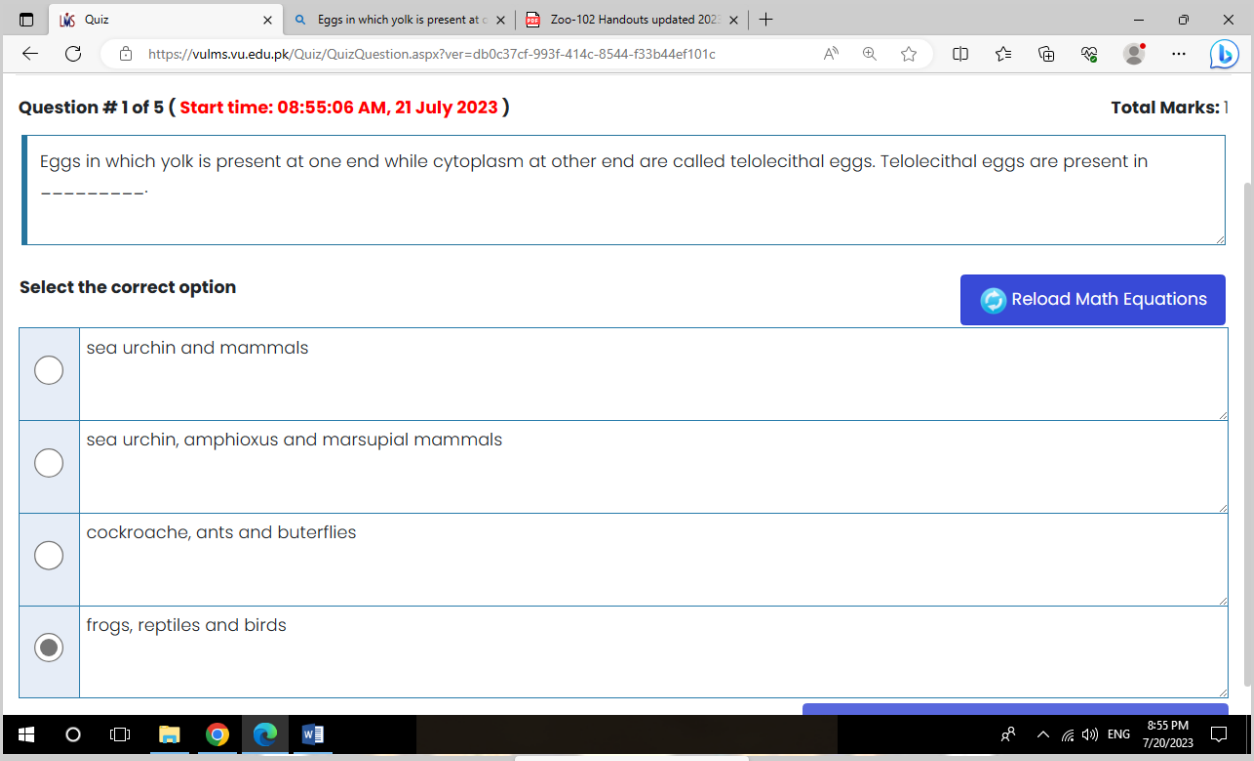
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Quiz | Eggs in which yolk is present at | Zoo-102 Handouts updated 202

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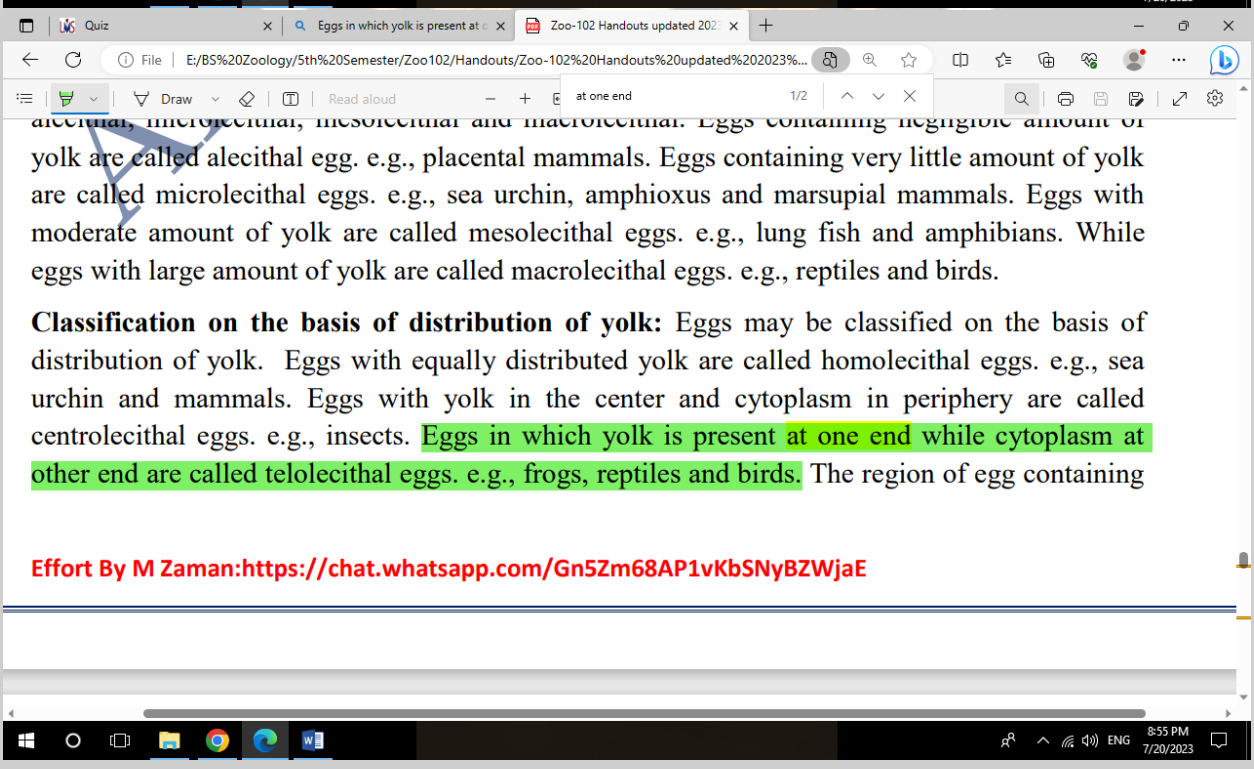
Question # 1 of 5 (Start time: 08:55:06 AM, 21 July 2023) **Total Marks:** 1

Eggs in which yolk is present at one end while cytoplasm at other end are called telolecithal eggs. Telolecithal eggs are present in _____.

Select the correct option

Reload Math Equations

- ○ sea urchin and mammals
- ○ sea urchin, amphioxus and marsupial mammals
- ○ cockroache, ants and buterflies
- ◉ frogs, reptiles and birds



Quiz | Eggs in which yolk is present at | Zoo-102 Handouts updated 202

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Draw | Read aloud | at one end 1/2

alecithal, microlecithal, mesolecithal and macrolecithal. Eggs containing negligible amount of yolk are called alecithal egg. e.g., placental mammals. Eggs containing very little amount of yolk are called microlecithal eggs. e.g., sea urchin, amphioxus and marsupial mammals. Eggs with moderate amount of yolk are called mesolecithal eggs. e.g., lung fish and amphibians. While eggs with large amount of yolk are called macrolecithal eggs. e.g., reptiles and birds.

Classification on the basis of distribution of yolk: Eggs may be classified on the basis of distribution of yolk. Eggs with equally distributed yolk are called homolecithal eggs. e.g., sea urchin and mammals. Eggs with yolk in the center and cytoplasm in periphery are called centrolecithal eggs. e.g., insects. Eggs in which yolk is present at one end while cytoplasm at other end are called telolecithal eggs. e.g., frogs, reptiles and birds. The region of egg containing

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Quiz | _____ clevage occur in the er | Zoo-102 Handouts updated 202

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Question # 2 of 5 (Start time: 08:55:38 AM, 21 July 2023) **Total Marks:** 1

_____ clevage occur in the embryo of nematodes.

Select the correct option

Reload Math Equations

- ○ indeterminate clevage
- ○ meroblastic clevage
- ◉ holoblastic clevage
- ○ determinate clevage

Quiz | _____ clevage occur in the er | Zoo-102 Handouts updated 202

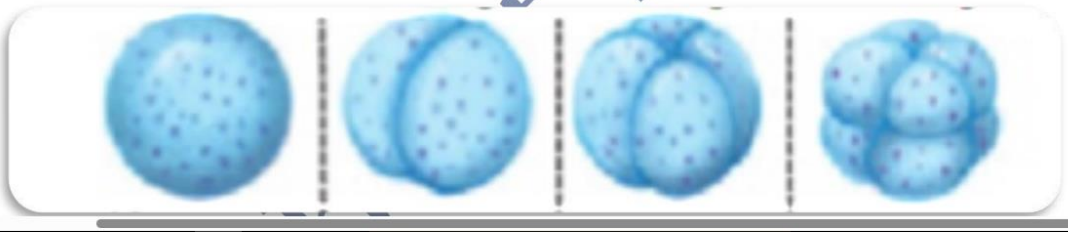
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Draw | Read aloud | Nematode 2/2

of embryo divide at the same time this is called **synchrony**. As development proceeds, all cells don't divide simultaneously. This is called **asynchrony**. Eggs with evenly distributed yolk have cleavage pattern that result uniform size blastomeres. Egg with unevenly distributed yolk result in unequal size blastomeres. There are following important patterns of cleavage:

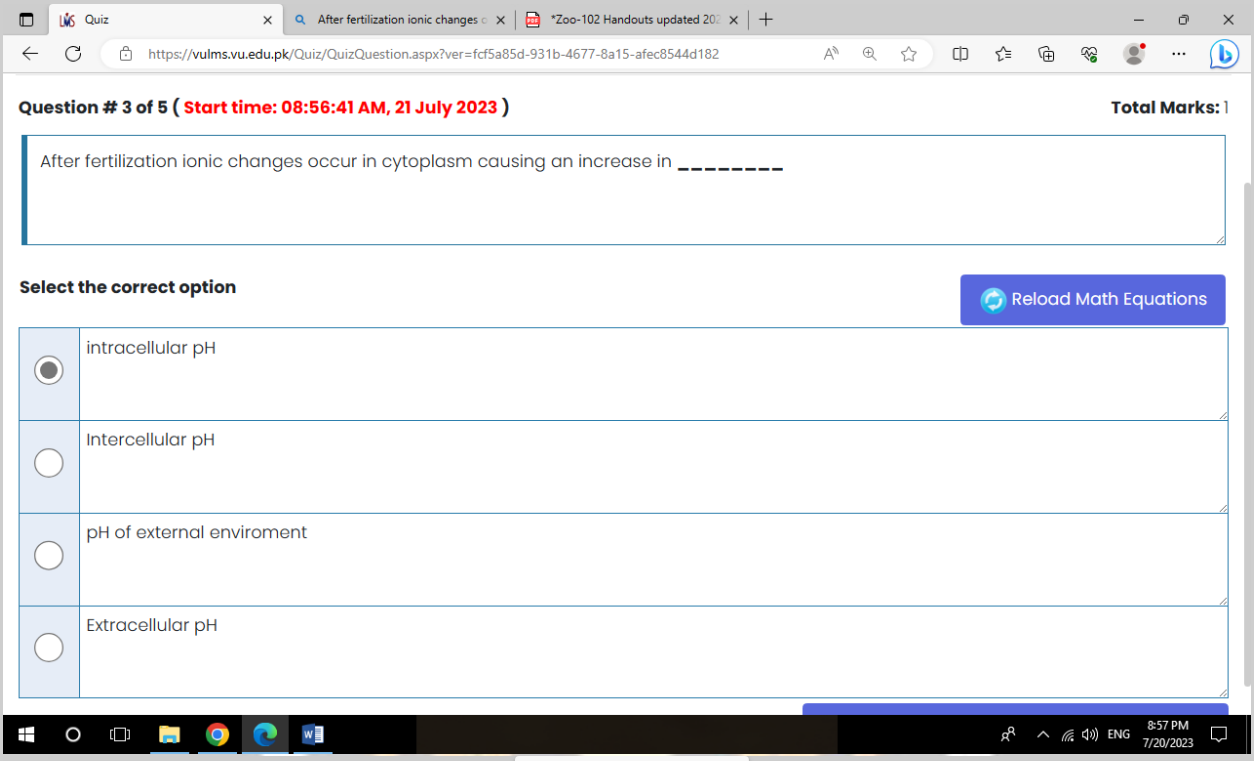
a. Holoblastic Cleavage

The pattern of cleavage in which egg is completely divided into two cells. Egg contain little or moderate amount of yolk. Cleavage in fertilized eggs of Amphibians, mammals, echinoderms, flatworms and nematodes.



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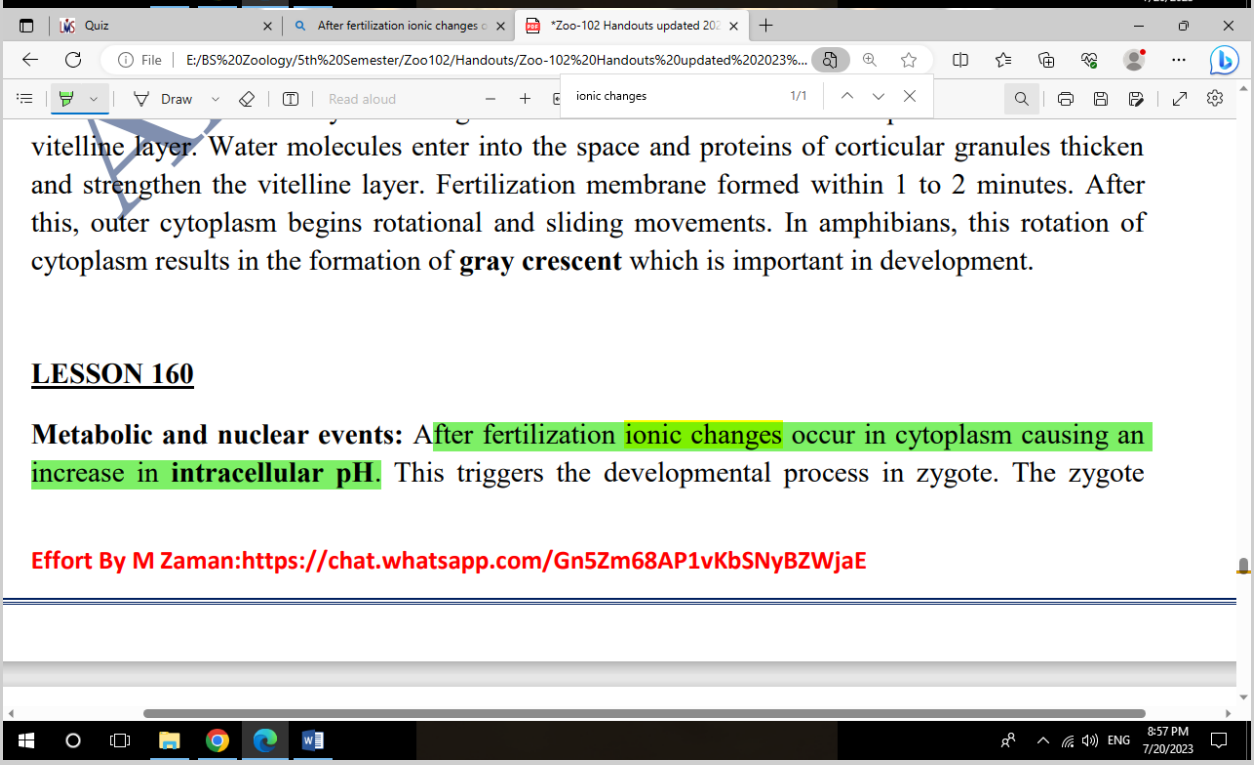
Question # 3 of 5 (Start time: 08:56:41 AM, 21 July 2023) Total Marks: 1

After fertilization ionic changes occur in cytoplasm causing an increase in _____

Select the correct option

Reload Math Equations

- intracellular pH
- Intercellular pH
- pH of external enviroment
- Extracellular pH



vitelline layer. Water molecules enter into the space and proteins of corticular granules thicken and strengthen the vitelline layer. Fertilization membrane formed within 1 to 2 minutes. After this, outer cytoplasm begins rotational and sliding movements. In amphibians, this rotation of cytoplasm results in the formation of **gray crescent** which is important in development.

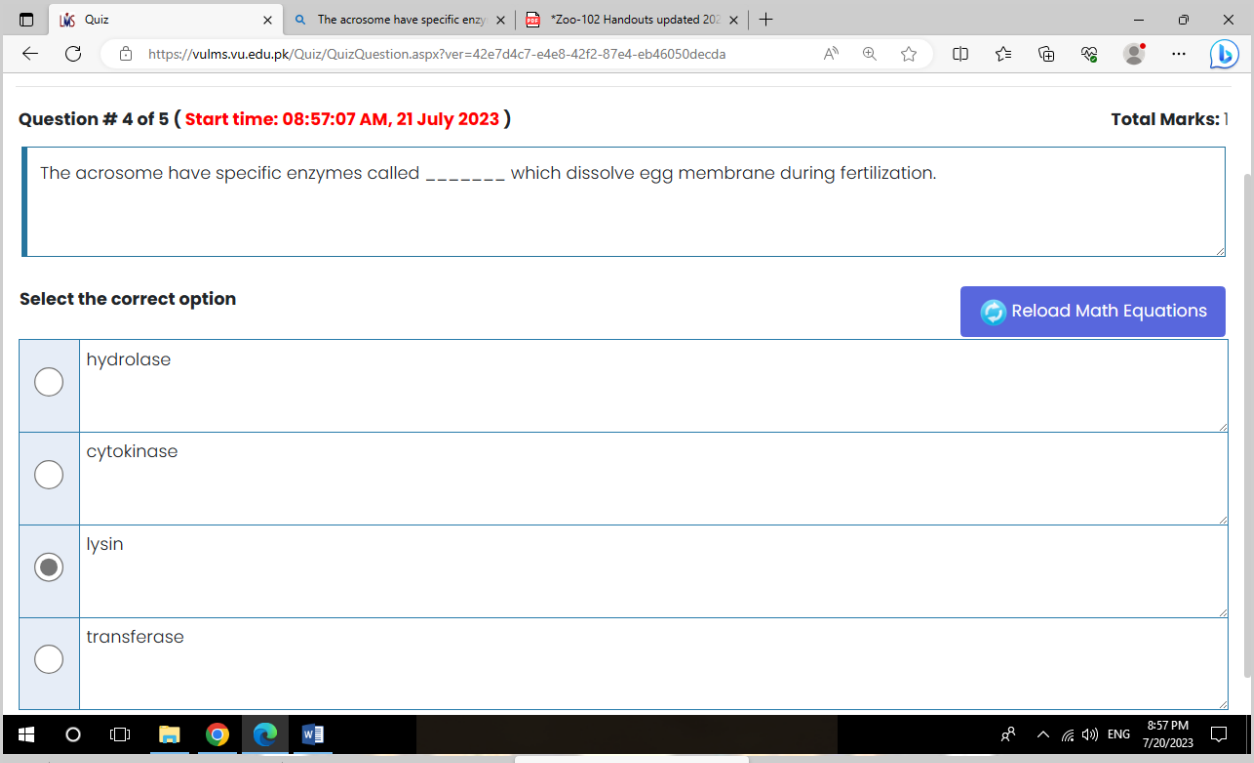
LESSON 160

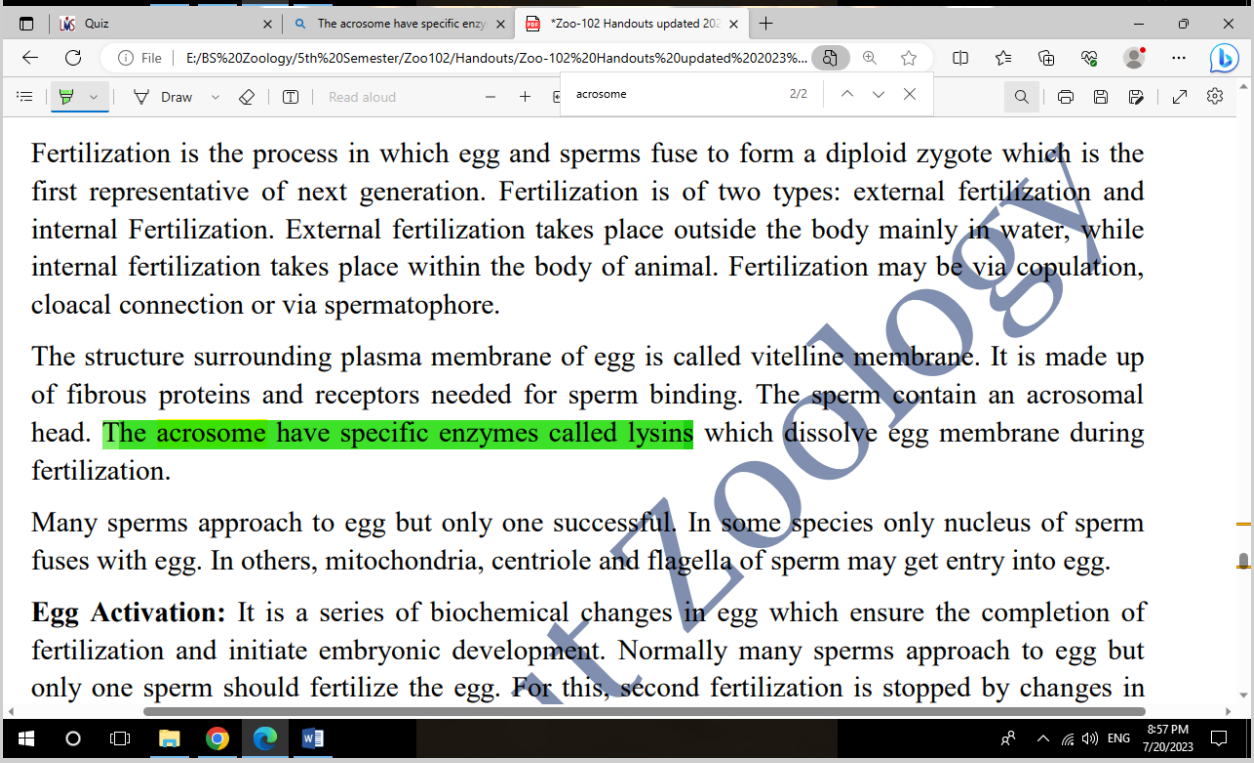
Metabolic and nuclear events: After fertilization ionic changes occur in cytoplasm causing an increase in **intracellular pH**. This triggers the developmental process in zygote. The zygote

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Fertilization is the process in which egg and sperms fuse to form a diploid zygote which is the first representative of next generation. Fertilization is of two types: external fertilization and internal Fertilization. External fertilization takes place outside the body mainly in water, while internal fertilization takes place within the body of animal. Fertilization may be via copulation, cloacal connection or via spermatophore.

The structure surrounding plasma membrane of egg is called vitelline membrane. It is made up of fibrous proteins and receptors needed for sperm binding. The sperm contain an acrosomal head. The acrosome have specific enzymes called lysins which dissolve egg membrane during fertilization.

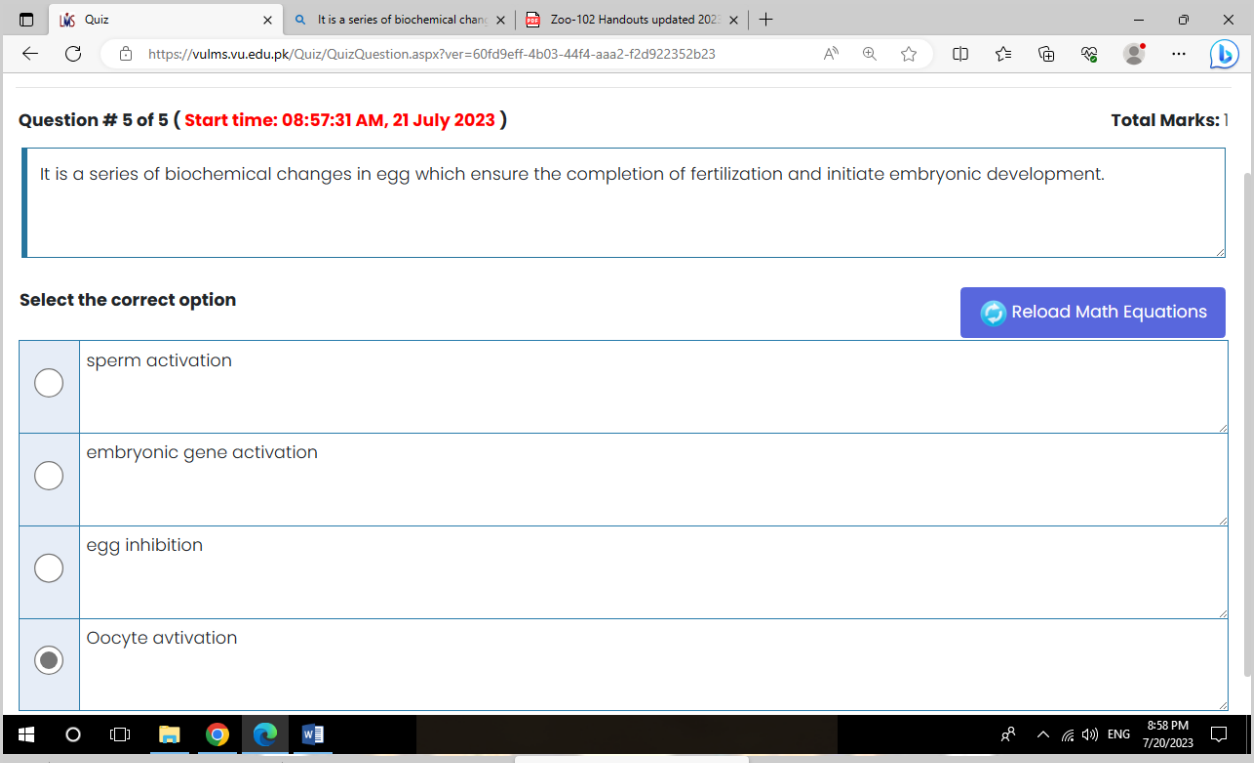
Many sperms approach to egg but only one successful. In some species only nucleus of sperm fuses with egg. In others, mitochondria, centriole and flagella of sperm may get entry into egg.

Egg Activation: It is a series of biochemical changes in egg which ensure the completion of fertilization and initiate embryonic development. Normally many sperms approach to egg but only one sperm should fertilize the egg. For this, second fertilization is stopped by changes in

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Quiz | It is a series of biochemical chan | Zoo-102 Handouts updated 202

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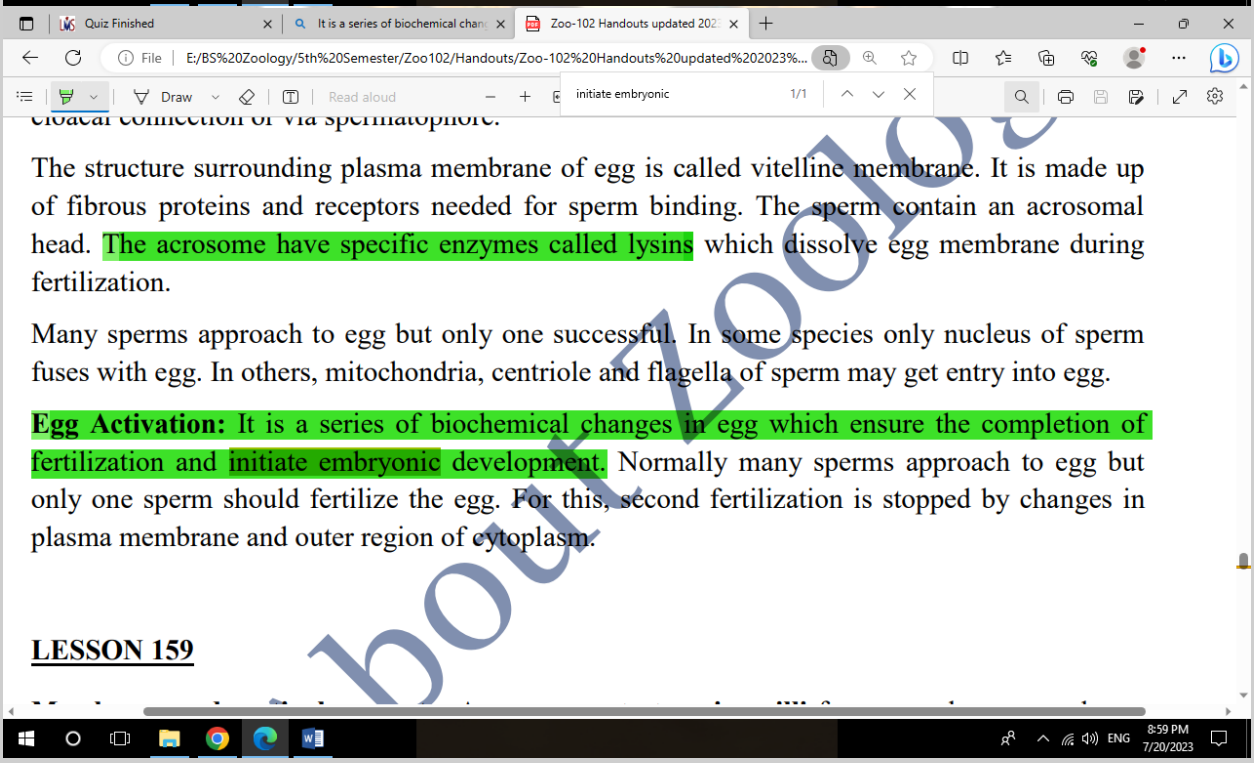
Question # 5 of 5 (Start time: 08:57:31 AM, 21 July 2023) **Total Marks:** 1

It is a series of biochemical changes in egg which ensure the completion of fertilization and initiate embryonic development.

Select the correct option

Reload Math Equations

- ○ sperm activation
- ○ embryonic gene activation
- ○ egg inhibition
- ◉ Oocyte avtivation



Quiz Finished | It is a series of biochemical chan | Zoo-102 Handouts updated 202

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Draw | Read aloud | initiate embryonic 1/1

cloacal connection or via spermatophore.

The structure surrounding plasma membrane of egg is called vitelline membrane. It is made up of fibrous proteins and receptors needed for sperm binding. The sperm contain an acrosomal head. The acrosome have specific enzymes called lysins which dissolve egg membrane during fertilization.

Many sperms approach to egg but only one successful. In some species only nucleus of sperm fuses with egg. In others, mitochondria, centriole and flagella of sperm may get entry into egg.

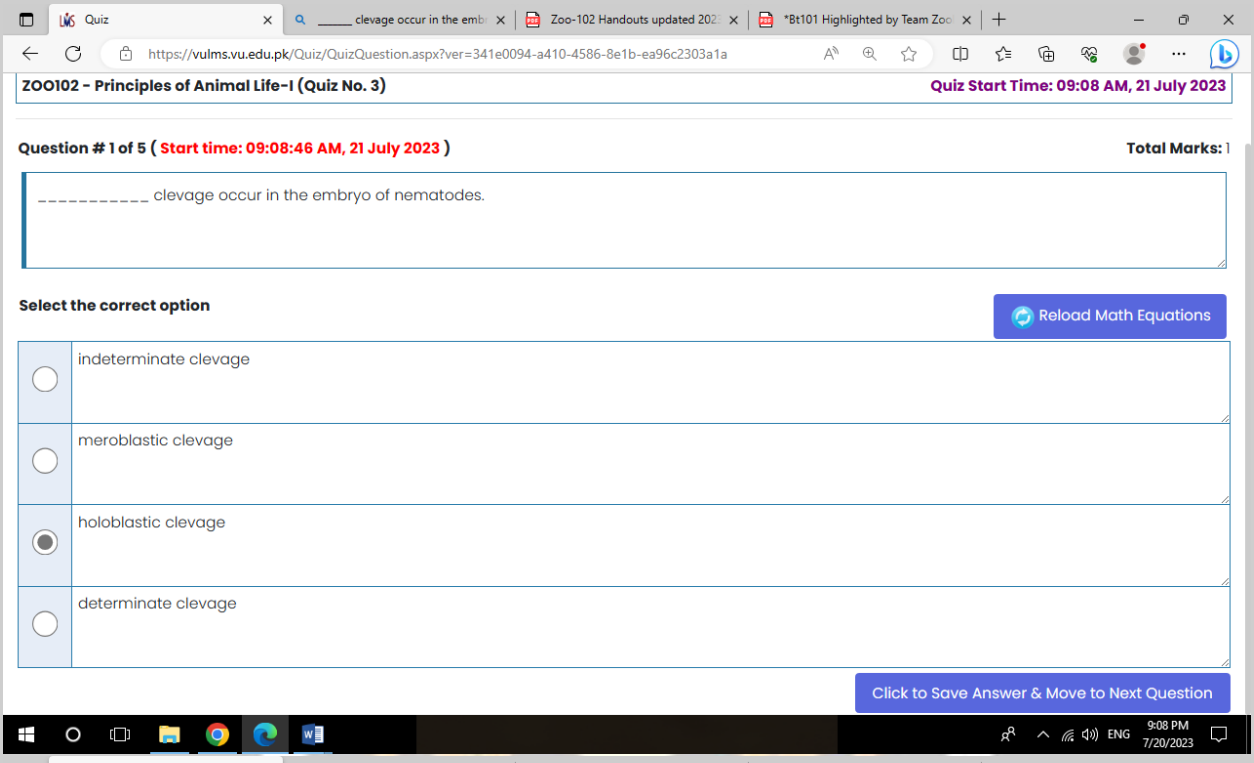
Egg Activation: It is a series of biochemical changes in egg which ensure the completion of fertilization and initiate embryonic development. Normally many sperms approach to egg but only one sperm should fertilize the egg. For this, second fertilization is stopped by changes in plasma membrane and outer region of cytoplasm.

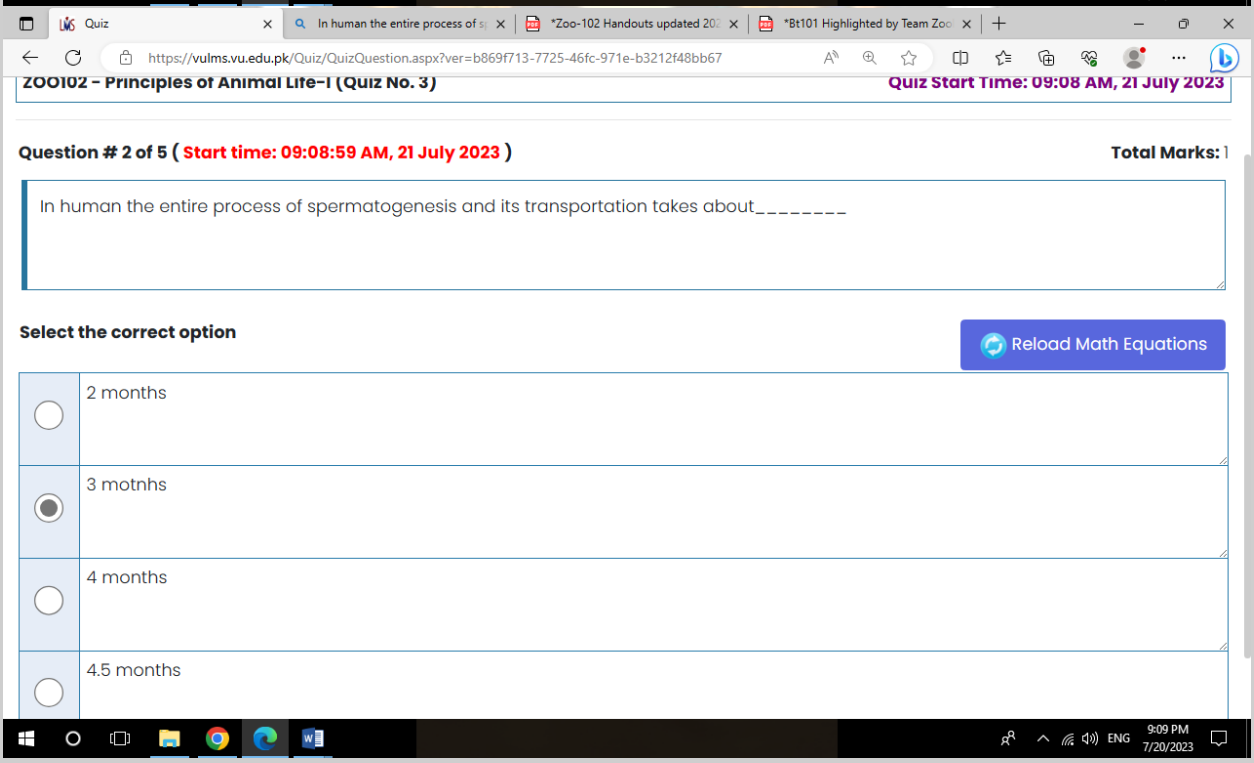
LESSON 159

bout Zoology

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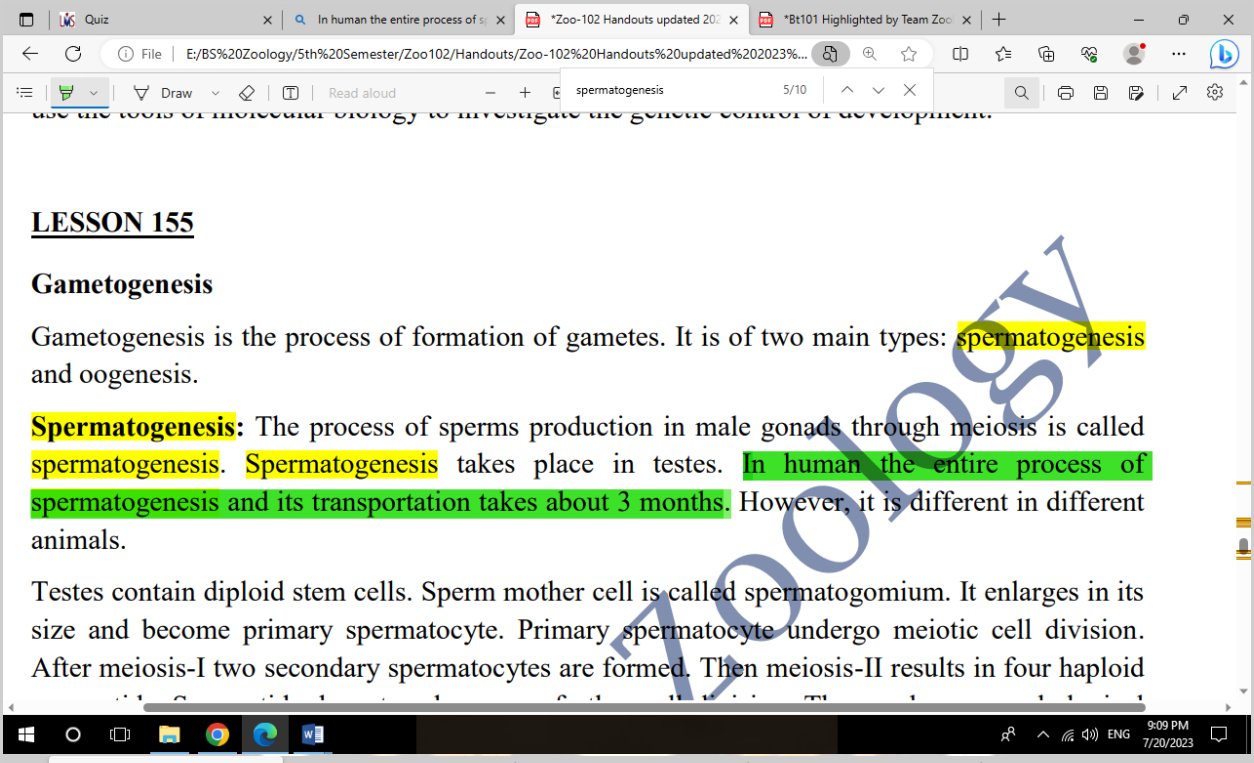
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LESSON 155

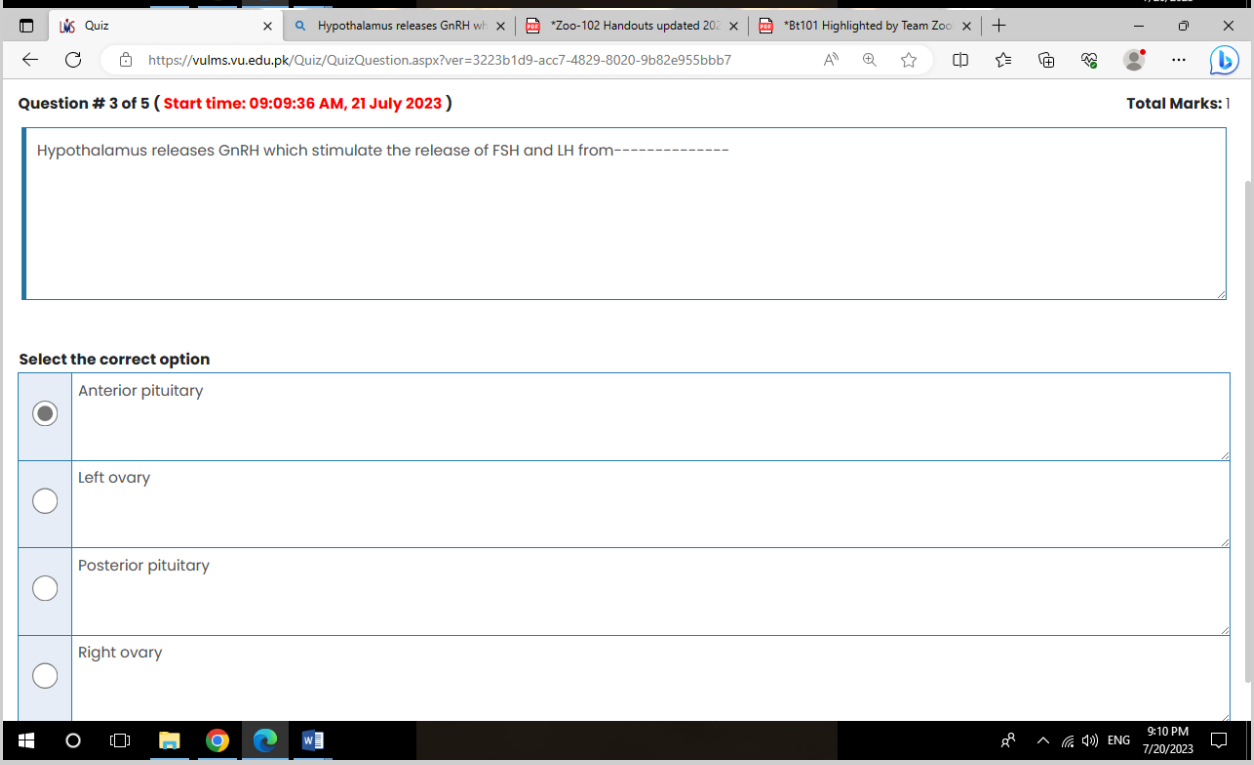
Gametogenesis

Gametogenesis is the process of formation of gametes. It is of two main types: spermatogenesis and oogenesis.

Spermatogenesis: The process of sperms production in male gonads through meiosis is called spermatogenesis. Spermatogenesis takes place in testes. In human the entire process of spermatogenesis and its transportation takes about 3 months. However, it is different in different animals.

Testes contain diploid stem cells. Sperm mother cell is called spermatogomium. It enlarges in its size and become primary spermatocyte. Primary spermatocyte undergo meiotic cell division. After meiosis-I two secondary spermatocytes are formed. Then meiosis-II results in four haploid

Zoology



Question # 3 of 5 (Start time: 09:09:36 AM, 21 July 2023) **Total Marks:** 1

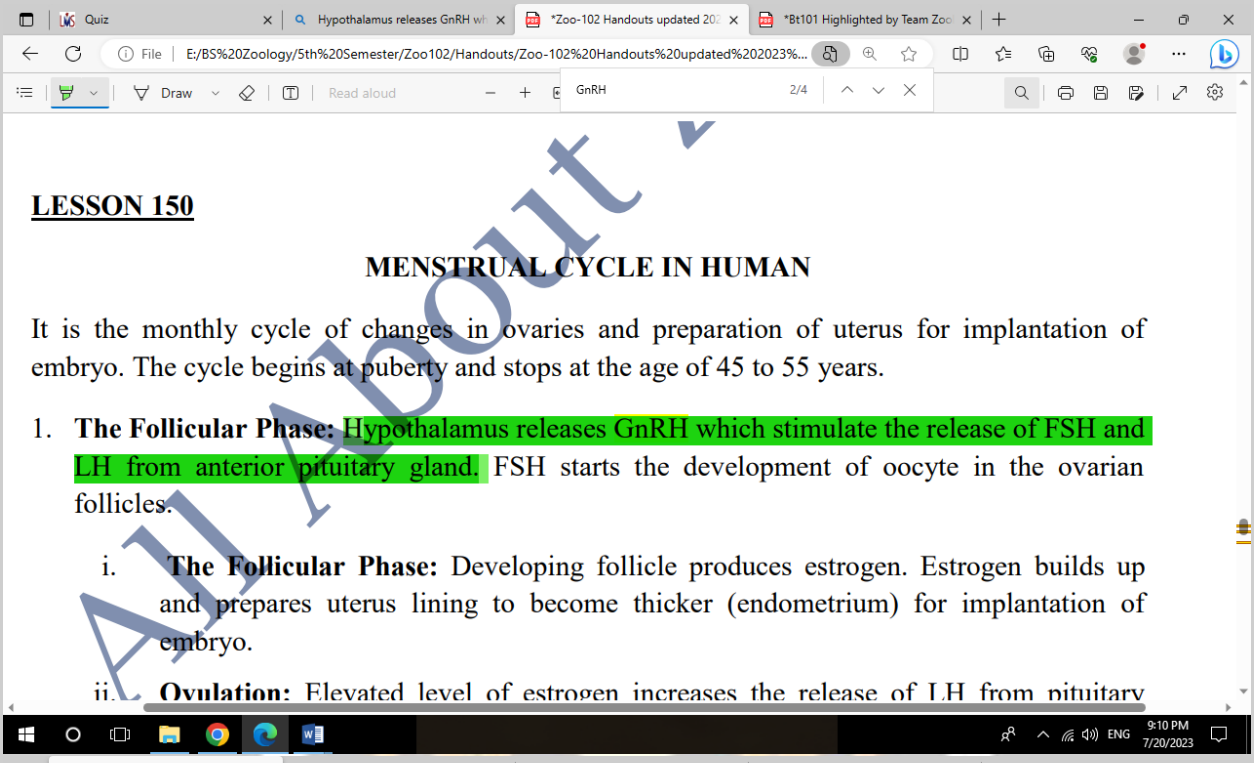
Hypothalamus releases GnRH which stimulate the release of FSH and LH from----------

Select the correct option

- (●) Anterior pituitary
- ( ) Left ovary
- ( ) Posterior pituitary
- ( ) Right ovary

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LESSON 150

MENSTRUAL CYCLE IN HUMAN

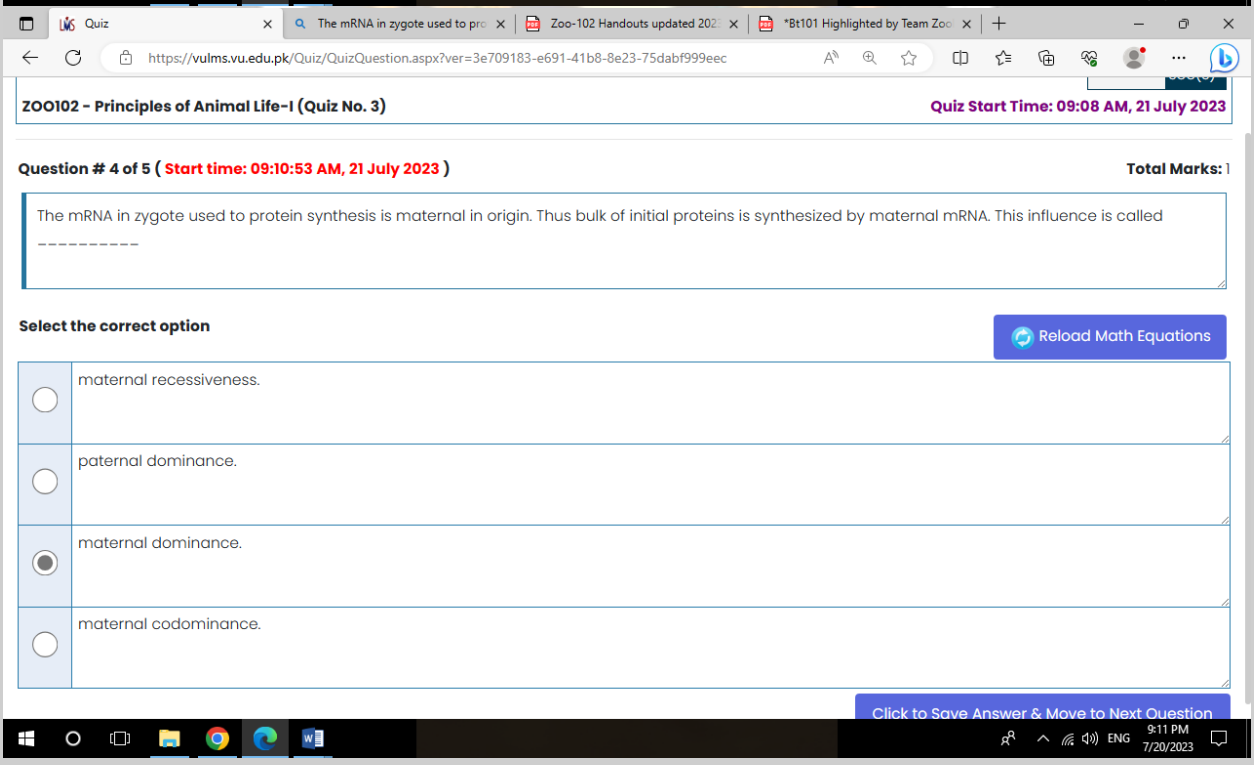
It is the monthly cycle of changes in ovaries and preparation of uterus for implantation of embryo. The cycle begins at puberty and stops at the age of 45 to 55 years.

1. **The Follicular Phase:** Hypothalamus releases GnRH which stimulate the release of FSH and LH from anterior pituitary gland. FSH starts the development of oocyte in the ovarian follicles.

   i. **The Follicular Phase:** Developing follicle produces estrogen. Estrogen builds up and prepares uterus lining to become thicker (endometrium) for implantation of embryo.

   ii. **Ovulation:** Elevated level of estrogen increases the release of LH from pituitary

All About



ZOO102 - Principles of Animal Life-I (Quiz No. 3) — Quiz Start Time: 09:08 AM, 21 July 2023

Question # 4 of 5 (Start time: 09:10:53 AM, 21 July 2023) — Total Marks: 1

The mRNA in zygote used to protein synthesis is maternal in origin. Thus bulk of initial proteins is synthesized by maternal mRNA. This influence is called _____

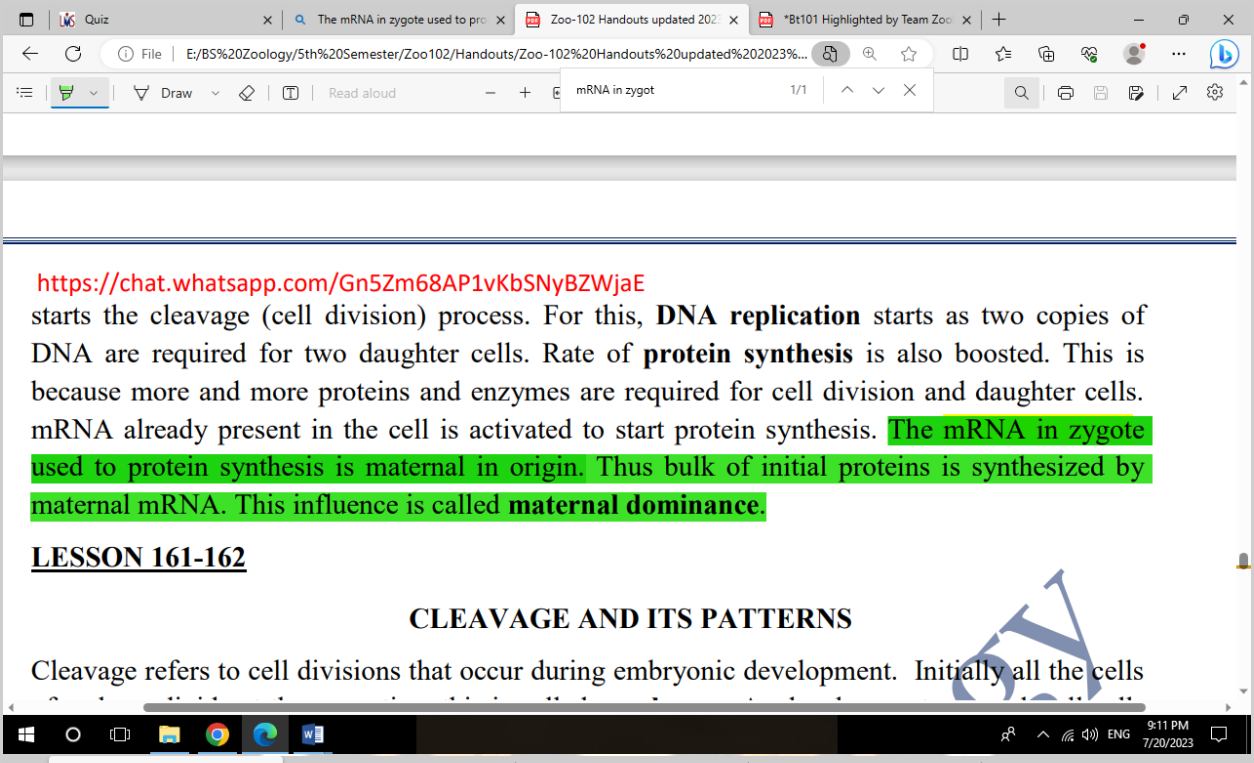
Select the correct option — Reload Math Equations

- maternal recessiveness.
- paternal dominance.
- maternal dominance.
- maternal codominance.

Click to Save Answer & Move to Next Question

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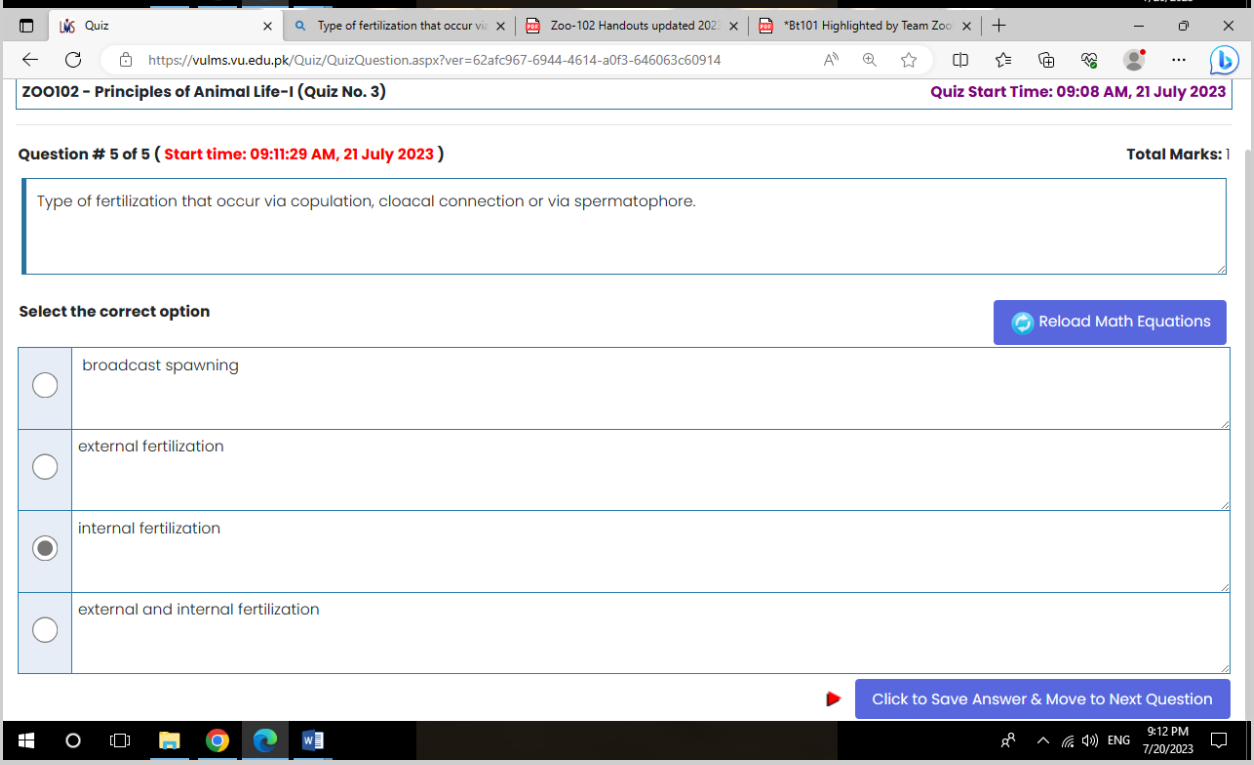
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starts the cleavage (cell division) process. For this, **DNA replication** starts as two copies of DNA are required for two daughter cells. Rate of **protein synthesis** is also boosted. This is because more and more proteins and enzymes are required for cell division and daughter cells. mRNA already present in the cell is activated to start protein synthesis. The mRNA in zygote used to protein synthesis is maternal in origin. Thus bulk of initial proteins is synthesized by maternal mRNA. This influence is called **maternal dominance**.

LESSON 161-162

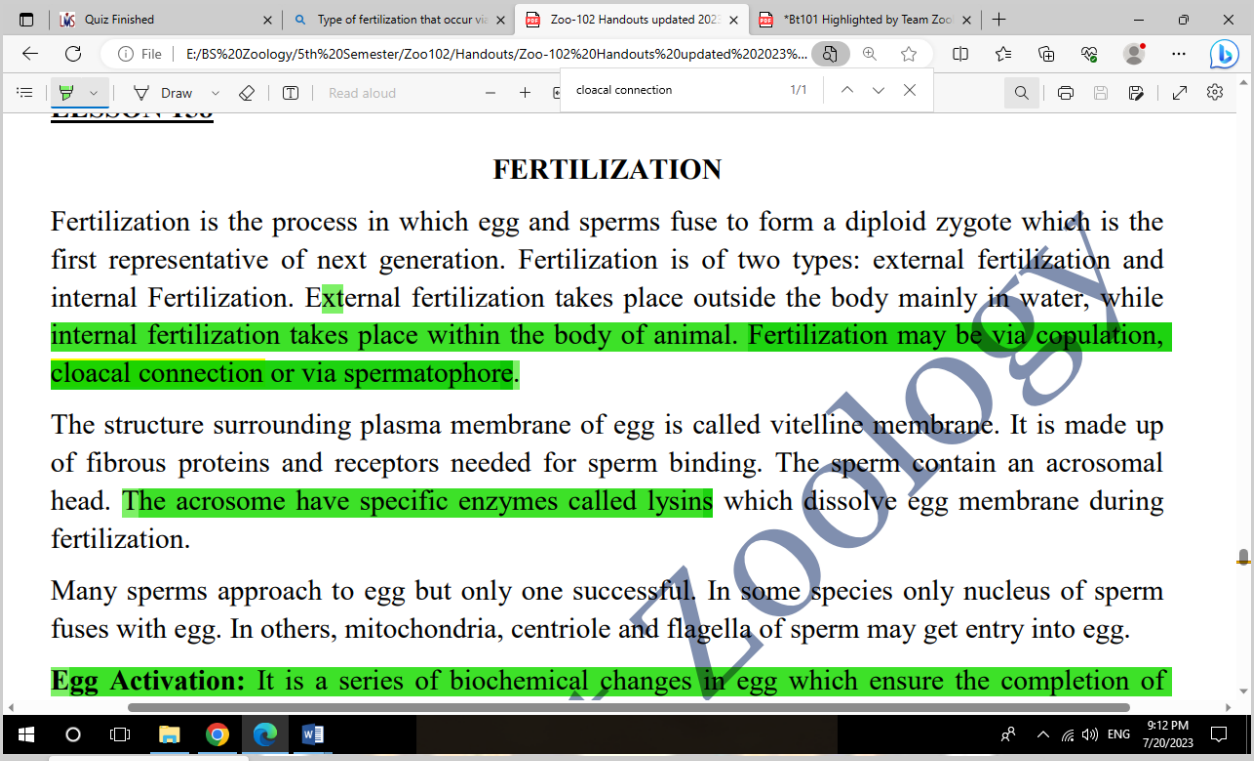
CLEAVAGE AND ITS PATTERNS

Cleavage refers to cell divisions that occur during embryonic development. Initially all the cells



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FERTILIZATION

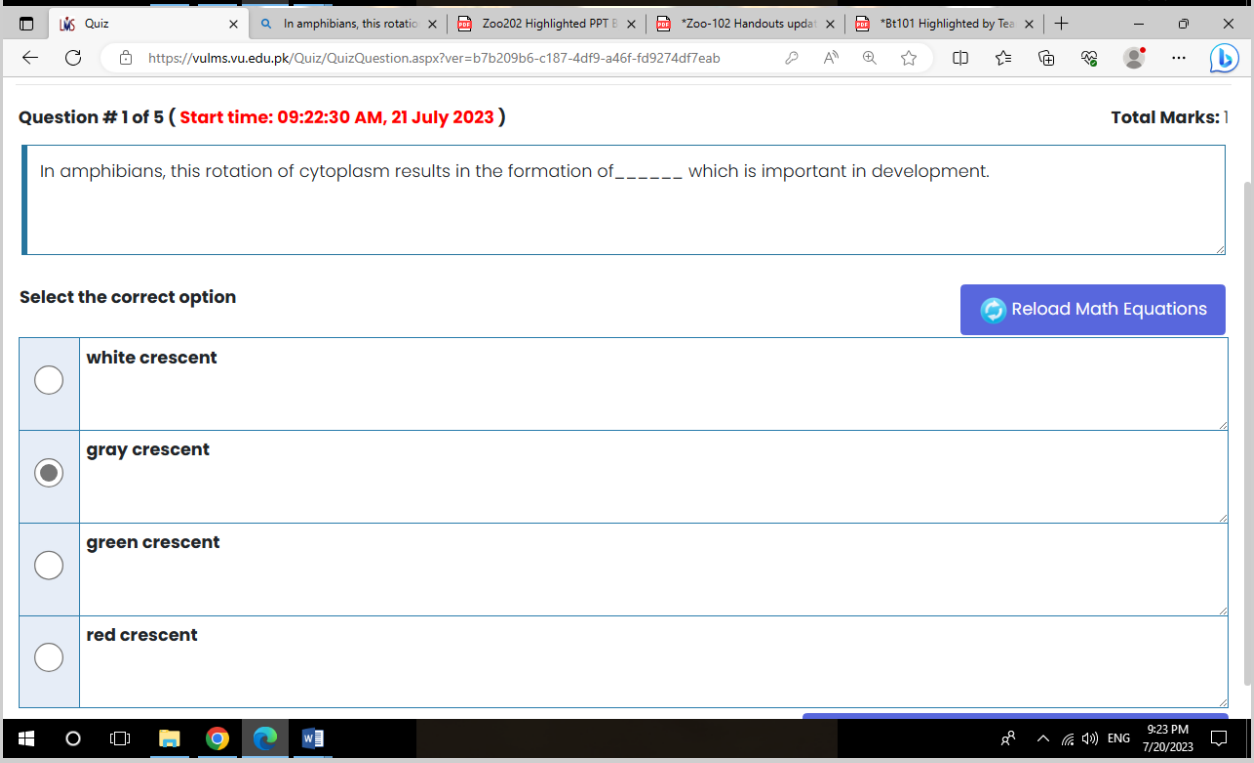
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The structure surrounding plasma membrane of egg is called vitelline membrane. It is made up of fibrous proteins and receptors needed for sperm binding. The sperm contain an acrosomal head. The acrosome have specific enzymes called lysins which dissolve egg membrane during fertilization.

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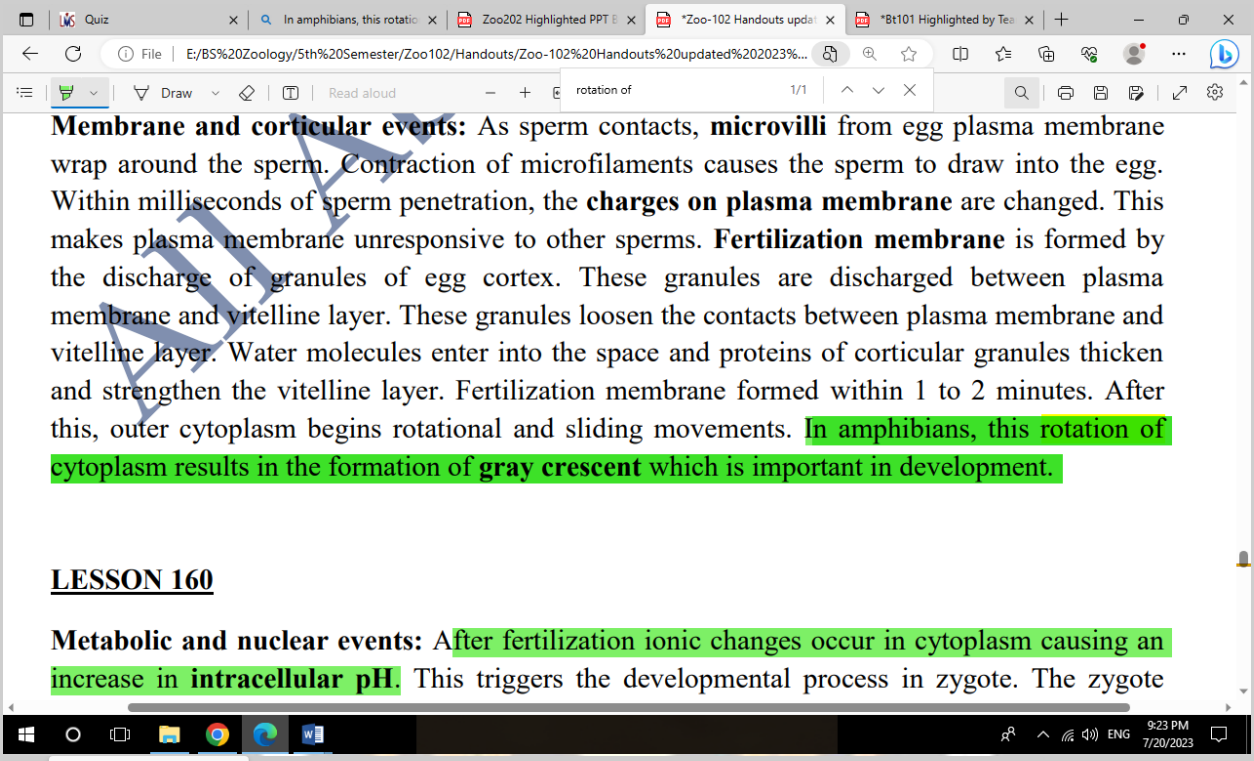
Egg Activation: It is a series of biochemical changes in egg which ensure the completion of

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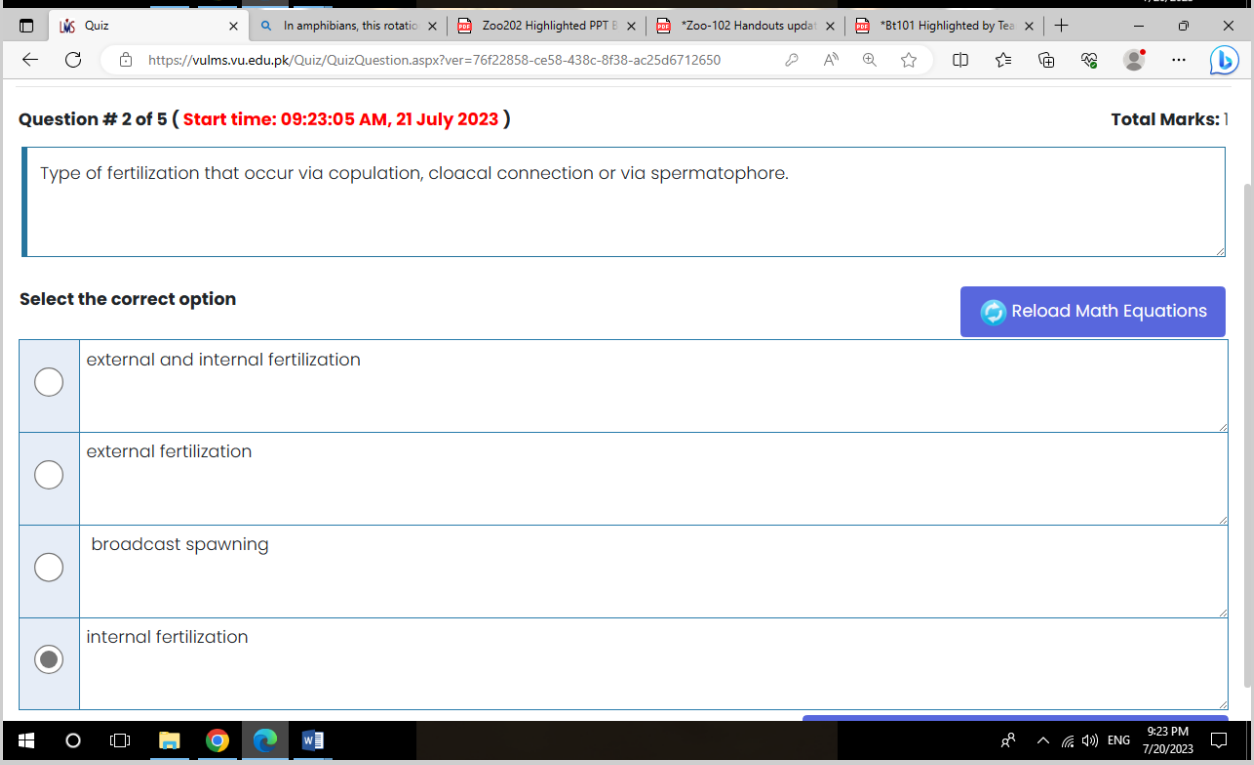
Zoology Legends



Membrane and corticular events: As sperm contacts, **microvilli** from egg plasma membrane wrap around the sperm. Contraction of microfilaments causes the sperm to draw into the egg. Within milliseconds of sperm penetration, the **charges on plasma membrane** are changed. This makes plasma membrane unresponsive to other sperms. **Fertilization membrane** is formed by the discharge of granules of egg cortex. These granules are discharged between plasma membrane and vitelline layer. These granules loosen the contacts between plasma membrane and vitelline layer. Water molecules enter into the space and proteins of corticular granules thicken and strengthen the vitelline layer. Fertilization membrane formed within 1 to 2 minutes. After this, outer cytoplasm begins rotational and sliding movements. In amphibians, this rotation of cytoplasm results in the formation of **gray crescent** which is important in development.

LESSON 160

Metabolic and nuclear events: After fertilization ionic changes occur in cytoplasm causing an increase in **intracellular pH**. This triggers the developmental process in zygote. The zygote



Question # 2 of 5 (Start time: 09:23:05 AM, 21 July 2023) **Total Marks:** 1

Type of fertilization that occur via copulation, cloacal connection or via spermatophore.

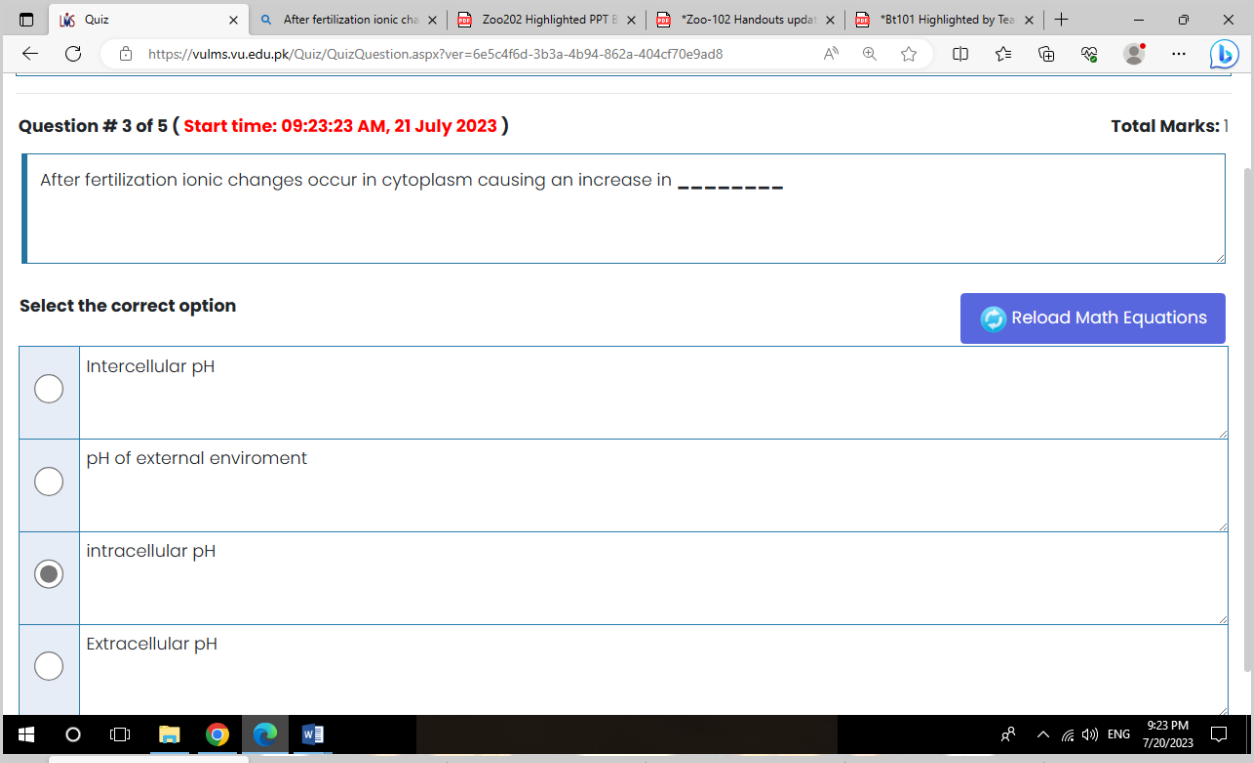
Select the correct option

Reload Math Equations

- external and internal fertilization
- external fertilization
- broadcast spawning
- internal fertilization

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Quiz | After fertilization ionic cha | Zoo202 Highlighted PPT B | *Zoo-102 Handouts updat | *Bt101 Highlighted by Tea

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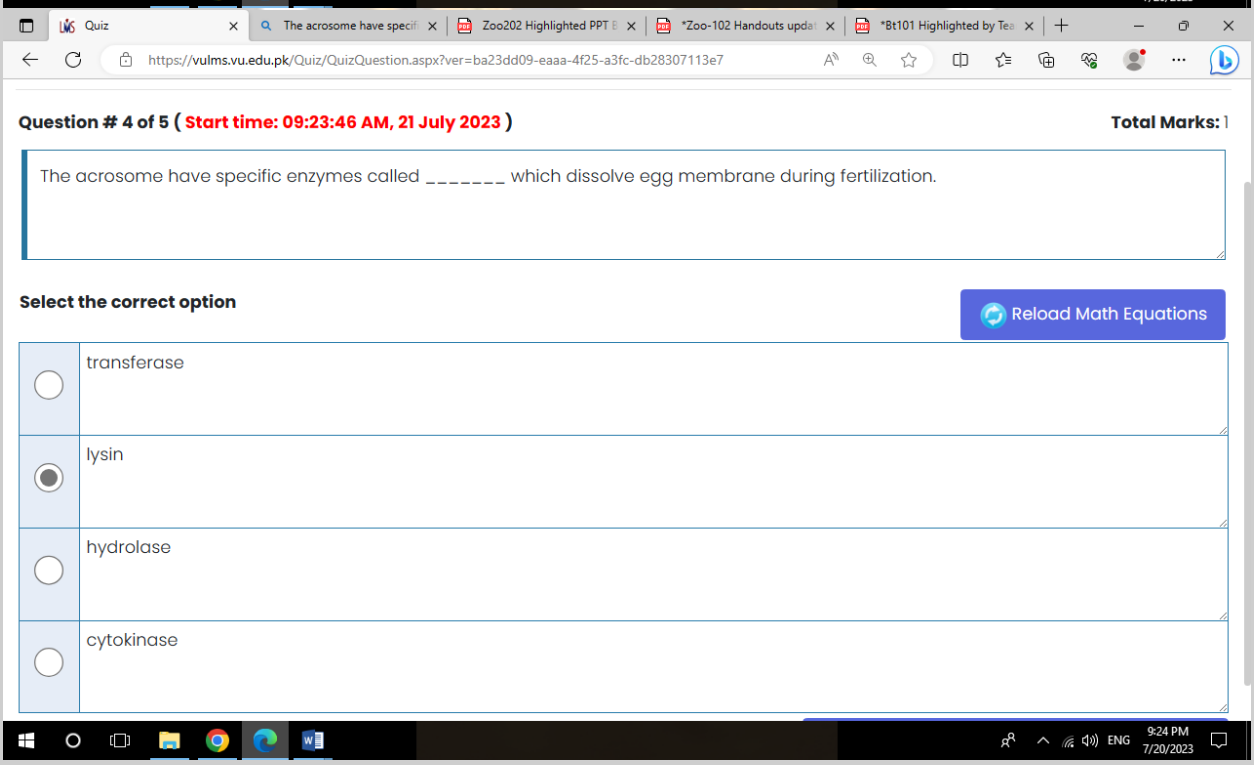
Question # 3 of 5 (Start time: 09:23:23 AM, 21 July 2023) **Total Marks:** 1

After fertilization ionic changes occur in cytoplasm causing an increase in _____

Select the correct option

Reload Math Equations

- ○ Intercellular pH
- ○ pH of external enviroment
- ● intracellular pH
- ○ Extracellular pH



Quiz | The acrosome have specif | Zoo202 Highlighted PPT B | *Zoo-102 Handouts updat | *Bt101 Highlighted by Tea

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Question # 4 of 5 (Start time: 09:23:46 AM, 21 July 2023) **Total Marks:** 1

The acrosome have specific enzymes called _____ which dissolve egg membrane during fertilization.

Select the correct option

Reload Math Equations

- ○ transferase
- ● lysin
- ○ hydrolase
- ○ cytokinase

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Quiz | A mature sperm consists | Zoo202 Highlighted PPT | *Zoo-102 Handouts updat | *Bt101 Highlighted by Tea

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Question # 5 of 5 (Start time: 09:24:17 AM, 21 July 2023) **Total Marks:** 1

A mature sperm consists of head which contain ---------- nucleus

Select the correct option

- ○ Diploid
- ○ Triploid
- ◉ Haploid
- ○ Teraploid

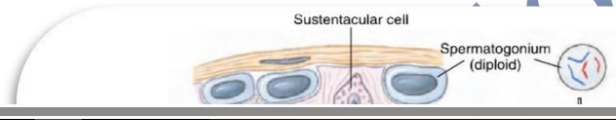
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Quiz Finished | A mature sperm consists | Zoo202 Highlighted PPT | *Zoo-102 Handouts updat | *Bt101 Highlighted by Tea

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Draw | Read aloud | f head 1/1

contain about 800 seminiferous tubules. Seminiferous tubules walls contain two types of cells: Spermatogenic cells and Sustentacular cells. There are clusters of interstitial cells between seminiferous tubules. Thousands of sperms are produced in each second in seminiferous tubules. The process of sperm production is called spermatogenesis. Spermatogenic cells are also called spermatogonia (singl. spermatogomium) are present in the wall of seminiferous tubules. A Spermatogomium increases in size and becomes primary spermatocyte. Meiosis-I results in two secondary spermatocytes. After meiosis-II four non-motile haploid spermatids are formed. Spermatid undergo morphological changes and are transformed into motile sperms. A mature sperm consists of head which contain haploid nucleus and acrosome cap having acrosin, midpiece which have mitochondria and tail that contains microtubules for whip like movements. Sustentacular cells provide nourishment to sperms and maintain fluid environment while



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Muhammad Zaman

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Remember me in you Precious Prayer

Jazak Allah

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